ASSIGNMENT 4

Q9: Shopping Cart Total

Instructions: Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

Solution

```
class Cart {
  constructor() { this.items = []; }
  addItem(name, price, quantity) {
    if (price < 0 || quantity < 1) throw new Error("Invalid price or quantity.");
    this.items.push({ name, price, quantity });
  }
  getTotal() {
    return this.items.reduce((sum, item) => sum + item.price * item.quantity, 0);
  }
  applyCoupon(code) {
    const match = /^(SAVE|DISC)(\d{1,2})$/.exec(code);
    if (!match) return { valid: false, message: "Use a coupon like SAVE20 or DISC10." };
    const percent = Number(match[2]);
    if (percent < 1 || percent > 100) return { valid: false, message: "Percent must be 1–100." };
    const total = this.getTotal();
    return { valid: true, discountPercent: percent, finalTotal: total * (1 - percent / 100) };
  }
}
const cart = new Cart();
cart.addItem("Notebook", 100, 2);
cart.addItem("Pen", 20, 3);
console.log("Cart total:", cart.getTotal()); // 260
const result = cart.applyCoupon("SAVE20");
console.log(result.valid ? `Final total: ■${result.finalTotal}` : result.message); // ■208
```

Expected result / testing

SAVE20 applies a 20% discount to ■260, making the final total ■208. Invalid coupon formats show an error.